

stored $[N, M]$

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$D_s = B = 8$: one bit a dimension

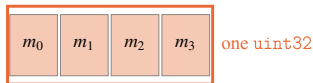
c_{00}	c_{01}	c_{02}	c_{03}	c_{04}	x_0
c_{10}	c_{11}	c_{12}	c_{13}	c_{14}	x_1
c_{20}	c_{21}	c_{22}	c_{23}	c_{24}	x_2
c_{30}	c_{31}	c_{32}	c_{33}	c_{34}	x_3

transpose
→

c_{00}	c_{10}	c_{20}	c_{30}	m_0
c_{01}	c_{11}	c_{21}	c_{31}	m_1
c_{02}	c_{12}	c_{22}	c_{32}	m_2
c_{03}	c_{13}	c_{23}	c_{33}	m_3
c_{04}	c_{14}	c_{24}	c_{34}	m_4

one candidate is a row;
the warp strides by M

one subspace is a row;
the warp reads 32 adjacent bytes



4 subspaces $\times$ 8 bits = 32 dimensions

byte layout



4 loads, 4 addresses, 4 loop iterations -- and the sectors it fetches are already the right ones

packed



1 load, 1 address, 1 iteration;
the same bytes move

(a) candidate-major $\rightarrow$ subspace-major

(b) four subspaces in one load: fewer instructions,
not fewer bytes (+30% to +48%)